

Oil Slick Identification in Marine Radar Image Using HOG, Random Forest, and PSO

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Abstract—Marine oil spills have become a significant threat to ocean environments and ecosystems. The development of effective marine oil spill identification technology is crucial for emergency authorities to enhance response strategies. In this letter, a marine oil spill detection method based on the histogram of oriented gradient (HOG) features, random forest classifier, and particle swarm optimization (PSO) algorithm was proposed. First, row vector convolution and mean filtering were adopted to extract and smooth the co-frequency interferences. Then, binarization processing and median filtering were used to eliminate speckles. After that, gray correction and contrast enhancement model were performed to enhance the overall oil slick features. Next, HOG features and random forest classifier were utilized to extract effective oil spill regions. Finally, the PSO algorithm was introduced to iteratively optimize the adaptive dual-threshold for segmenting the real oil slicks. This approach enables accurate and efficient detection of oil spills, offering a scientific foundation for responding to offshore oil spill incidents.

Index Terms—Machine learning, oil pollution, radar detection.

I. INTRODUCTION

WITH the rapid development of industrialized countries around the world, the exploitation of marine resources is increasing, which in turn has led to significant growth

in oil extraction and shipping [1]. Overpressure, mechanical failure, pipeline corrosion, and ship collisions are all potential causes of unanticipated oil leaks during the exploration, production, and transportation processes [2]. Oil spill has tremendous adverse effects on the marine living resources and ecosystems [3]. Therefore, accurate and rapid marine oil spill detection is of great significance to ecological resources protection and human society development.

Marine X-band radar has become an indispensable technical means in ocean safety supervision because of its real time, stable, convenient, and economic advantages. Shipborne radar is adaptable to multiple platforms, including shore, ship, and platform. It can also efficiently undertake large-scale oil spill monitoring and early warning missions in bad weather circumstances [4]. Meantime, its high-light target detection range can reach 72 nmi or even further. The oil spill detection distance can reach more than 1 nmi under the condition of fine effective wave echo images on the sea surface [5]. The marine X-band radar allows continuous monitoring of the target area. This capability is particularly helpful in oil leak detection in the early stages of a ship [6]. In 1991, Atanassov et al. [7] successfully demonstrated that X-band marine radar can be used for marine oil spill detection by the property of oil slick suppression of ocean wave echoes. Zhu et al. [8] proposed a power attenuation correction method to solve the uneven distribution in echo information of X-band marine radar, which proved the validity of the extraction method for oil slicks.

In this letter, an oil spill detection method based on the histogram of oriented gradient (HOG) features, random forest classifier, and PSO algorithm was proposed. After image preprocessing, HOG features, feature weighted dimensionality reduction, and random forest classifier were used to extract the effective oil spill regions. Afterward, the particle swarm optimization (PSO) algorithm was used to calculate the adaptive dual threshold for the segmentation of real and suspected oil films. Finally, the real ones were further extracted to ensure more accurate identification.

II. METHODOLOGY

A. Data Preprocessing

The data preprocessing process was shown in Fig. 1. First, the original image of X-band marine radar in the polar coordinate system was converted into Cartesian coordinate system. Then, the row vector convolution, gray threshold

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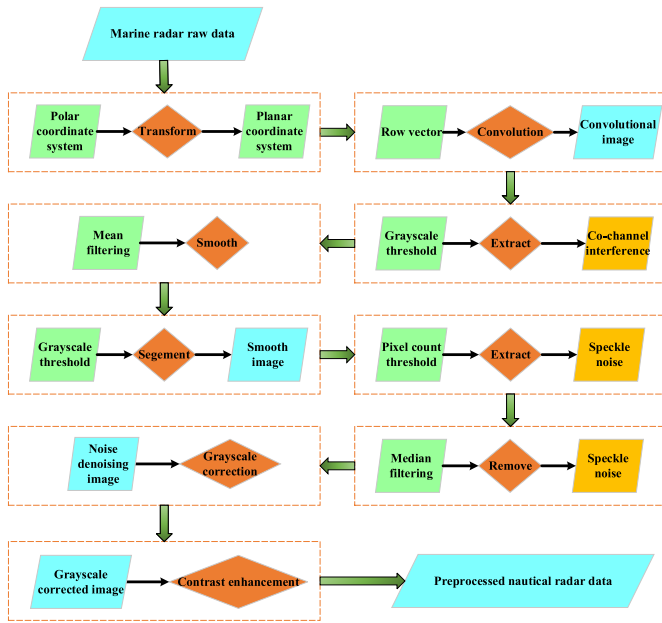


Fig. 1. Preprocessing of raw experimental data.

(“100”), and mean filter were performed to smooth co-channel interference. After that, the speckles were removed by using a pixel region threshold (“200”) and median filter. Finally, the gray correction and contrast enhancement model were performed to enhance the contrast of the oil slick features. Last, the contrast enhancement image was cut into tiles of 32×32 pixels.

B. Weighted Dimensionality Reduction of HOG

Texture features are widely used in image processing [9], [10], while morphological features are less commonly used in oil spill image processing. HOG is a descriptive operator that can detect the edge features of targets in an image. It is particularly effective in object classification and recognition, which works by calculating the gradient direction distribution of specific points in the image [11]. After the image preprocessing, the weighted dimensionality reduction of HOG feature extraction method was as follows.

- 1) The Gamma correction method was used to normalize each marine radar preprocessing tile $I_i(x, y)$.
- 2) The gradient $G(x, y)$ and orientations $\theta(x, y)$ of each pixel in each tile were calculated as follows:

$$G(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2} \quad (1)$$

$$\theta(x, y) = \arctan \frac{G_y(x, y)}{G_x(x, y)} \quad (2)$$

where $G_x(x, y)$ was the horizontal gradient and $G_y(x, y)$ was the vertical gradient.

- 3) The tiles were divided into small cells. The size of cell was 16×16 pixels here.
- 4) The histograms of gradient orientations for each cell were computed. In this letter, each tile was divided into four cells. Then, the gradient orientations of each cell were equally divided into nine bins (360° in total). In this

way, each cell produced nine features h_i , and four cells produced a total of 36 features.

- 5) It was proposed to use variance weights on HOG features to achieve the effect of feature dimensionality reduction. First, the variance σ_i of each cell was calculated. Then, the weight coefficient W_i for each cell was calculated using the following:

$$W_i = \frac{\sigma_i}{\sum_{i=1}^m \sigma_i} \quad (3)$$

where m represented the number of cells ($m = 4$). The features of each cell in each direction were fused to as v_j

$$v'_j = \sum_{i=1}^m W_i v_j \quad (4)$$

where v_j represents the histogram gradient in the j th direction of the i th cell.

C. Random Forest

Random forest is an effective ensemble learning method that consists of multiple decision trees and combines strategies, such as bootstrap sampling, random selection of features, and voting or averaging to reduce the risk of overfitting. The random forest can improve prediction performance and provide feature importance assessment for tasks, such as object classification and identification [12], [13]. The number of trees was set to “100”, and the minimum number of samples required for leaf nodes was set to “5” here.

D. PSO

PSO is a population-based stochastic evolution algorithm developed for continuous nonlinear function optimization problems [14]. In this algorithm, the foraging behavior of birds is simulated, the particle swarm is initialized and assigned velocity and position attributes, and then, the motion state of the particles is continuously updated according to the guidance of the individual and global optimal solutions. It is easy to understand and implement, with high computational efficiency and good adaptability. PSO can be used for image segmentation and feature extraction when processing radar or optical remote sensing images.

III. EXPERIMENTS

A. Pretreatment Process

Compared to longer wavelength S-band and L-band radars, X-band radars have higher operating frequencies (approximately between 8 and 12 GHz) and shorter wavelengths (usually between 2.5 and 3.75 cm). This enables it to provide higher resolution images and effectively detect small oil spill areas. So, the experimental platform used X-band marine radar. The experimental image was collected by the teaching–training ship *Yukun* of Dalian Maritime University during its cruise mission, as shown in Fig. 2. The size is 1024×1024 pixels. The detection range is 0.75 nmi (nm). The actual area represented by one pixel is 7.3441 m^2 . The specific pretreatment steps were shown as Fig. 3.

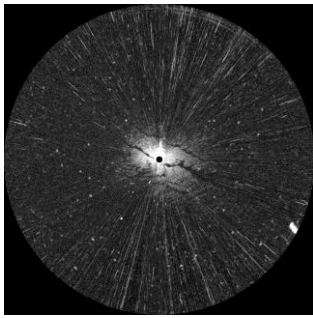


Fig. 2. Raw image of X-band marine radar in polar coordinate system.

B. Extraction of the Oil Spill Active Regions

According to the nine weighted characteristics of HOG, random forest was used to divide tiles into two categories: oil and oil-free. It should be noted that when using the random forest classification method, a part of the samples needs to be labeled first. Afterward, the classified tiles were fused, as shown in Fig. 4.

C. Preliminary Segmentation Result of Oil Slicks

The PSO algorithm was used to calculate the adaptive dual threshold. The real oil film (black targets) and suspected oil film (gray targets) were segmented, respectively, as shown in Fig. 5. The initial population size was set to “500”, the number of segmentation thresholds was set to “2”, the maximum number of iterations was set to “500”, the inertia weight coefficient was set to “0.8”, and both the self-learning factor and group learning factor were set to “0.5.”

D. Results

After a series of image processing and analysis processes, including preprocessing, HOG feature extraction, random forest classification, and PSO optimization, preliminary segmentation result was achieved. The real oil slick classification result was separately extracted, as shown in Fig. 6(a). Thereafter, the spots were removed as Fig. 6(b). Finally, the fine identified oil film targets were, respectively, marked in the denoised Cartesian coordinate system and polar coordinates, as shown in Fig. 6(c) and (d).

IV. DISCUSSION

A. Impacts of Ship Wakes and Ocean Waves on Marine Radar Oil Spill Monitoring

Due to the propeller power propulsion of the ship, the original shape of the ocean waves in the stern region is broken. Some image features formed by the ship wake (blue box region) were similar to the oil slick feature (red box region), as shown in Fig. 7.

In the 120°–240° region of the experimental data (Fig. 8), there is indeed a ribbon of oil film (blue polygonal region). However, because of the influence of the ship trajectory, many false-positive targets (purple polygon regions) were identified. Many reports have entirely removed targets identified in the wake region in marine radar images [15]. In order to give

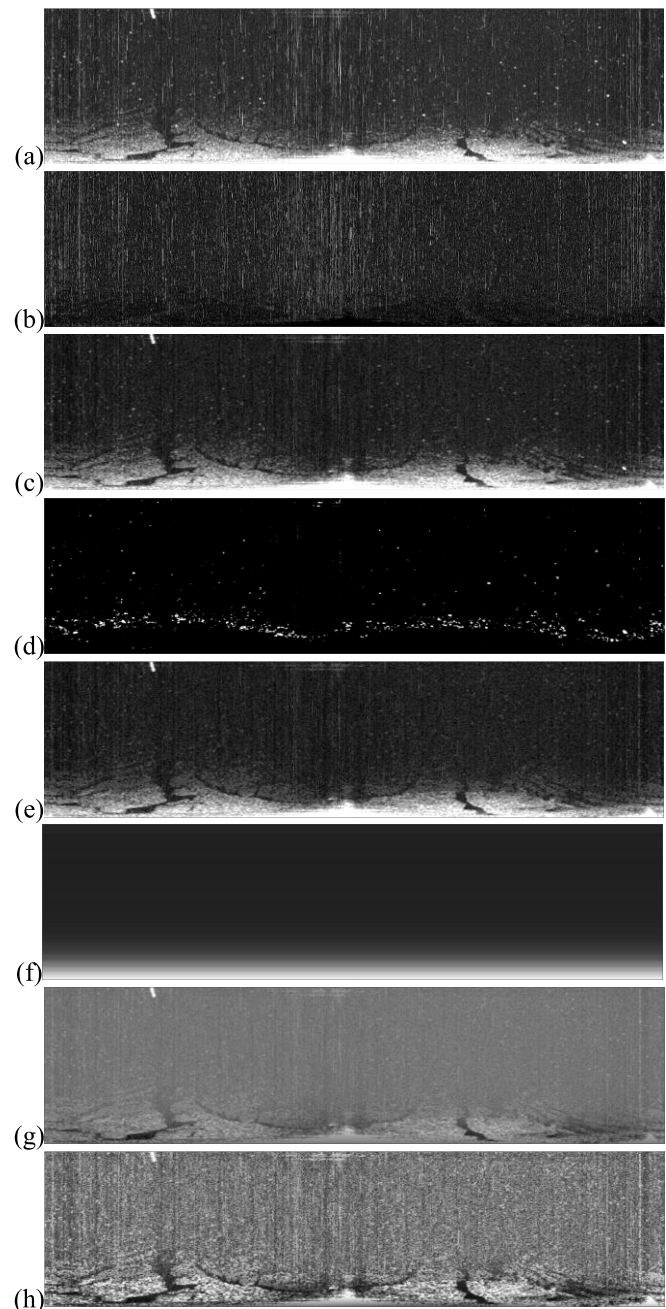


Fig. 3. Pretreatment processes. (a) Original image in Cartesian coordinate system. (b) Row vector convolution image. (c) Co-frequency interference suppression image. (d) Denoised image in Cartesian coordinate system. (e) Speckles extraction image. (f) Speckles removal image. (g) Gray correction matrix. (h) Contrast local enhancement image.

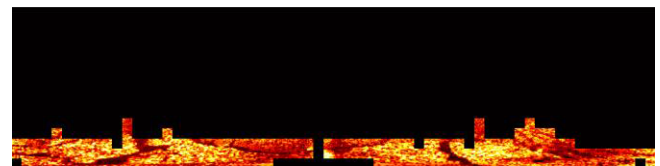


Fig. 4. Oil spill active regions extracted using random forest.

attention to both the automaticity and integrity of oil slick identification, some false positive targets identified in ship wake region were still existed. How to automatically retain



Fig. 5. Preliminary segmentation result by PSO.

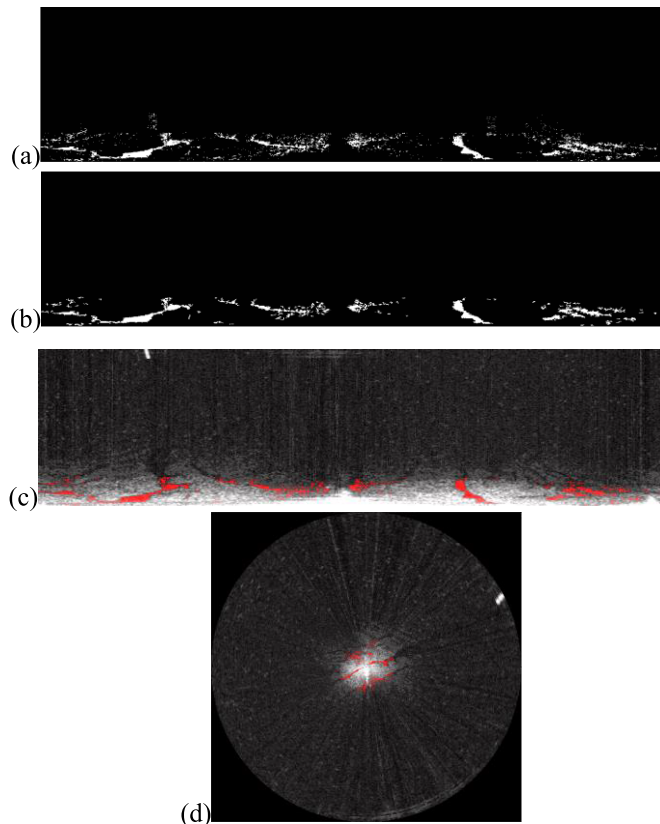


Fig. 6. Final results. (a) Real oil slick targets of preliminary segmentation. (b) Fine segmentation of real oil slicks. (c) Final result in Cartesian coordinate system. (d) Final result in polar coordinate system.

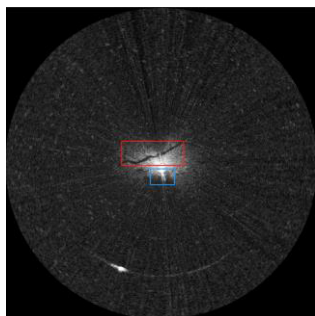


Fig. 7. Example of a false-positive oil slick image feature in the ship wake region.

all the real oil slick targets and effectively suppress the interference of ship wake will be an important research direction.

The marine radar oil spill monitoring equipment can accompany the clean-up ship to the accident site and has the advantage of real-time assistance in emergency response, but this technology is still constrained by the ocean wave conditions. When the waves are too large, the oil film has a weaker

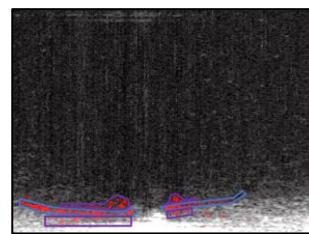


Fig. 8. Oil slick segmentation in 120–240 azimuth.

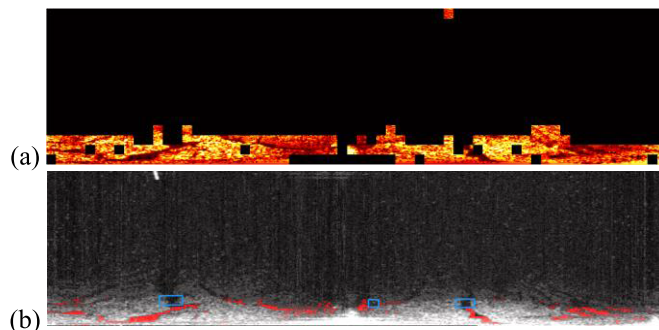


Fig. 9. Oil slick segmentation of HOG features without dimensionality reduction. (a) Oil spill region extraction. (b) Oil spill identification.

effect on suppressing the ocean waves. When the sea surface is too calm, the radar cannot retrieve wave echoes and cannot identify the oil film on the sea surface.

B. Effect of Feature Dimensionality Reduction

The 36 morphological features of HOG without weighted fusion were classified by random forest for oil film region detection, and the results were shown in Fig. 9(a). Afterward, it was classified by PSO algorithm for oil film analysis, and the real oil slick segmentations were shown in Fig. 9(b). The result of HOG features without dimensionality reduction missed some oil films in blue box regions, which indicated that too many feature dimensions reduced the classification accuracy of oil film activity regions.

V. CONCLUSION

The marine radar oil spill detection with image texture features, machine learning classification, and adaptive threshold segmentation is an extensive application method. In this letter, an oil spill identification method combining HOG features, random forest, and PSO algorithm is proposed to accurately segment the real oil slicks. This is a comprehensive application method of morphological features and machine learning. It can provide convenient and fast technical support for rapid response to oil spill accidents.

This method still involves some human intervention operations. With the continuous increase of oil spill data from marine radar, using deep learning networks for marine radar oil spill detection will become an effective approach.

REFERENCES

- [1] A. Galieriková and M. Materna, "World seaborne trade with oil: One of main cause for oil spills?" *Transp. Res. Proc.*, vol. 44, pp. 297–304, 2020.

- [2] K. Lee et al., "High-priority research needs for oil spills in Canada: Summary of a royal society expert panel report on the behaviour and environmental impacts of crude oil released into aqueous environments," in *Proc. 39th AMOP Tech. Seminar, Environ. Climate Change*, 2016, pp. 966–991.
- [3] H. Mei and Y. Yin, "Studies on marine oil spills and their ecological damage," *J. Ocean Univ. China*, vol. 8, no. 3, pp. 312–316, Sep. 2009.
- [4] J. Xu, X. Pan, B. Jia, X. Wu, P. Liu, and B. Li, "Oil spill detection using LBP feature and K-means clustering in shipborne radar image," *J. Mar. Sci. Eng.*, vol. 9, no. 1, p. 65, Jan. 2021.
- [5] B. Li et al., "Marine oil spill detection with X-band shipborne radar using GLCM, SVM and FCM," *Remote Sens.*, vol. 14, no. 15, p. 3715, Aug. 2022.
- [6] C.-M. Chiu et al., "Forecasting of oil-spill trajectories by using SCHISM and X-band radar," *Mar. Pollut. Bull.*, vol. 137, pp. 566–581, Dec. 2018.
- [7] V. Atanassov, L. Mladenov, R. Rangelov, and A. Savchenko, "Observation of oil slicks on the sea surface by using marine navigation radar," in *Proc. Remote Sens., Global Monit. Earth Manag. (IGARSS)*, 2005, pp. 1323–1326.
- [8] X. Zhu, Y. Li, H. Feng, B. Liu, and J. Xu, "Oil spill detection method using X-band marine radar imagery," *J. Appl. Remote Sens.*, vol. 9, no. 1, Nov. 2015, Art. no. 095985.
- [9] M. K. Kelishadrokhi, M. Ghattaei, and S. Fekri-Ershad, "Innovative local texture descriptor in joint of human-based color features for content-based image retrieval," *Signal, Image Video Process.*, vol. 17, no. 8, pp. 4009–4017, Nov. 2023.
- [10] S. Fekri-Ershad and M. F. Alsaffar, "Developing a tuned three-layer perceptron fed with trained deep convolutional neural networks for cervical cancer diagnosis," *Diagnostics*, vol. 13, no. 4, p. 686, Feb. 2023.
- [11] Md. G. Sarowar, M. A. Razzak, and M. A. A. Fuad, "HOG feature descriptor based PCA with SVM for efficient & accurate classification of objects in image," in *Proc. IEEE 9th Int. Conf. Adv. Comput. (IACC)*, Dec. 2019, pp. 171–175.
- [12] M. Belgiu and L. Drăguț, "Random forest in remote sensing: A review of applications and future directions," *ISPRS J. Photogramm. Remote Sens.*, vol. 114, pp. 24–31, Apr. 2016.
- [13] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–31, 2001.
- [14] R. Eberhart and J. Kennedy, "A new optimizer using particle swarm theory," in *Proc. 6th Int. Symp. Micro Machine Human Sci.*, 1995, pp. 39–43, doi: [10.1109/MHS.1995.494215](https://doi.org/10.1109/MHS.1995.494215).
- [15] J. Xu, H. Wang, C. Cui, B. Zhao, and B. Li, "Oil spill monitoring of shipborne radar image features using SVM and local adaptive threshold," *Algorithms*, vol. 13, no. 3, p. 69, Mar. 2020.